Quantum Computing and Technologies

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A century of Quantum Mechanics

Wolfgang Pauli, Heisenberg, Enrico Fermi, Archive, CERN

- On 9 July 1925, in a letter to Wolfgang Pauli, Werner Heisenberg revealed his new ideas, which were to revolutionise physics
- *All of my pitiful efforts are directed at completely killing off the concept of orbits – which, after all, cannot be observed – and replacing it with something more suitable,"* he explains in his letter to Pauli.
- Max Born, Pascual Jordan and Wolfgang Pauli himself helped turn Heisenberg's work into matrix mechanics, the first mature formulation of quantum theory.
- The era of quantum physics begins in 1925, with the discoveries of Schrodinger and Heisenberg. The seeds for these discoveries were planted by Planck, Einstein, Bohr, de Broglie, and others.

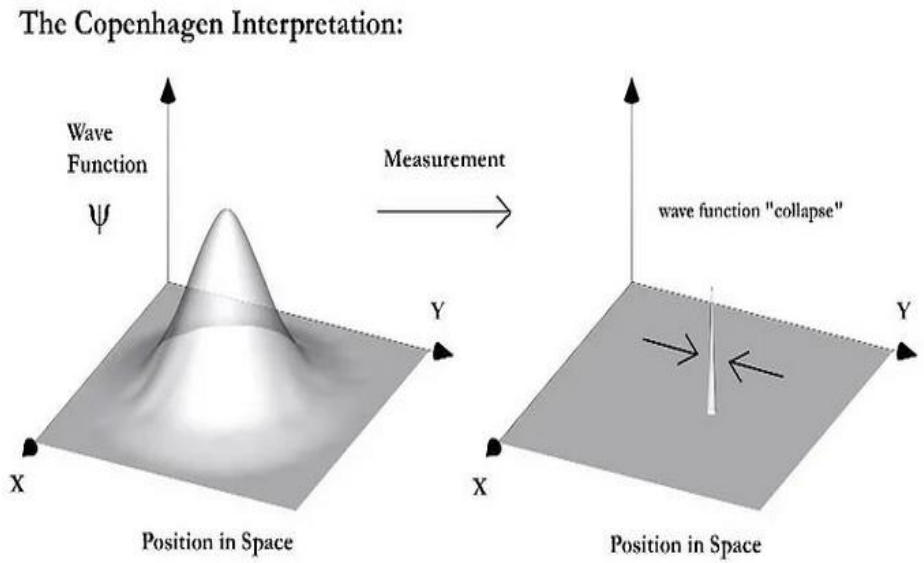
Key features of Quantum Mechanics

- Quantum mechanics is a linear theory.
- The signature equation Schrodinger equation is a linear equation for a quantity called the wavefunction Ψ
- Physical interpretation of Ψ was not clear to Erwin Schrodinger when he wrote the equation in 1925. It was Max Born, who suggested that the wavefunction encodes probabilities.
- The Schrodinger equation is a partial differential equation $i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$
- Hamiltonian is a linear operator $\hat{H}(a\Psi) = a\hat{H}\Psi, \quad \hat{H}(\Psi_1 + \Psi_2) = \hat{H}(\Psi_1) + \hat{H}(\Psi_2)$

Key features of Quantum Mechanics

- If Ψ_1 and Ψ_2 are solutions to the Schrodinger equation $\alpha\Psi_1 + \beta\Psi_2$, is also a solution.
- Complex numbers are sometimes useful in classical mechanics or Maxwell theory, they are not strictly needed. None of the dynamical variables, which correspond to measurable quantities, is a complex number.
- All measurements in physics result in real numbers
- Wavefunction is a complex number $\Psi \in \mathbb{C}$

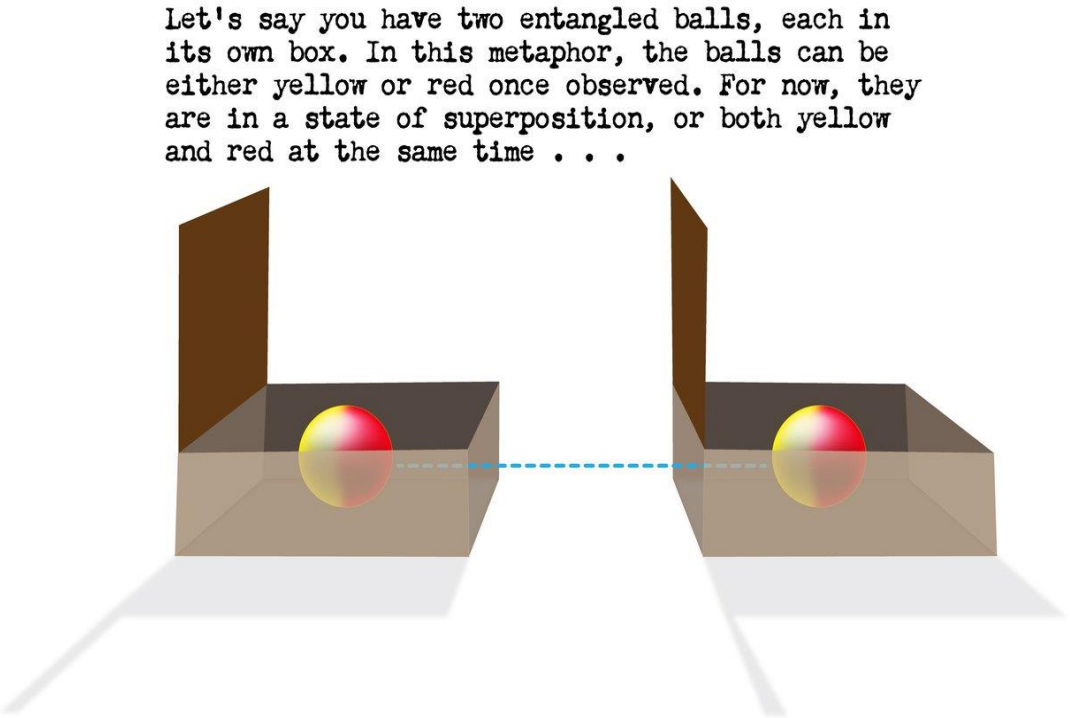
Key features of Quantum Mechanics



- Superposition: A wavefunction itself isn't very useful. But it can place the quantum information it holds into a state of superposition, which represents a combination of all possible configurations of the wavefunction. Groups of wavefunctions in superposition can create complex, multidimensional computational spaces.

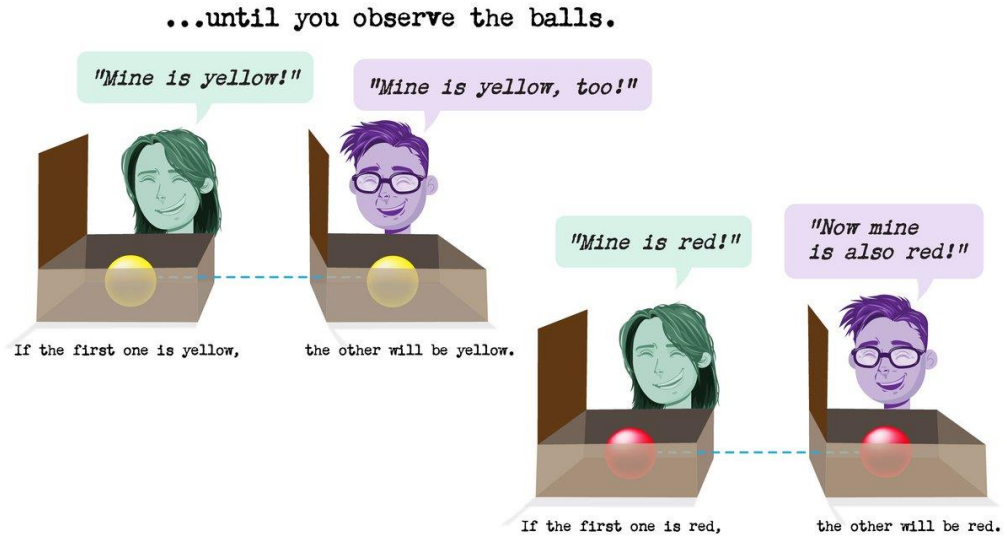
- When a quantum system is measured, its state collapses from a superposition of possibilities into a state.

Key features of Quantum Mechanics

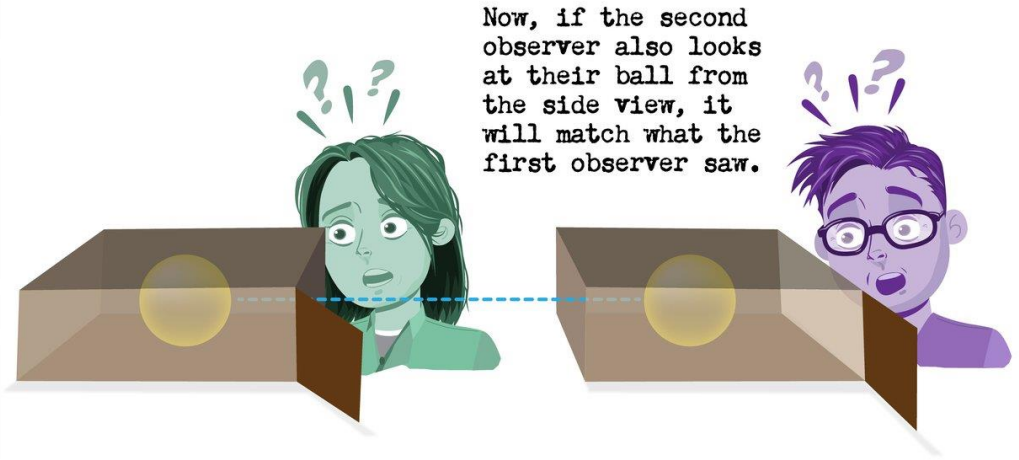


- Entanglement: Entanglement is the ability of a quantum state to correlate their state with other quantum state. Entangled systems are so intrinsically linked that when quantum processors measure a single entangled state, they can immediately determine information about other state in the entangled system.

Entanglement



The objects remain connected even over vast distances. Scientists think of entangled objects as really being a single object.



The balls are still entangled, but what the viewer sees depends on how they look at the ball. This is because the entangled information about color does not lie within any one ball but exists in the connection between the balls.

What is Quantum Computing?

- Quantum computing is an emergent field of computer science that harnesses the unique qualities of quantum mechanics to solve problems beyond the ability of even the most powerful classical computers.

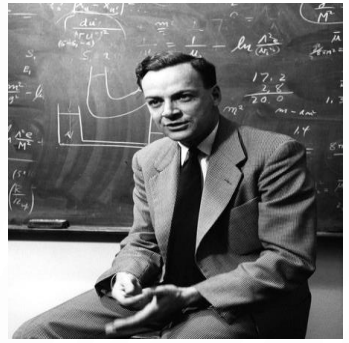
- The field of quantum computing includes a range of disciplines, including quantum hardware and quantum algorithms. While still in development, quantum technology will soon be able to solve complex problems that classical supercomputers can't solve (or can't solve fast enough).

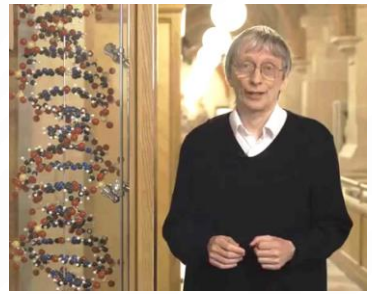
Classical computing versus Quantum computing

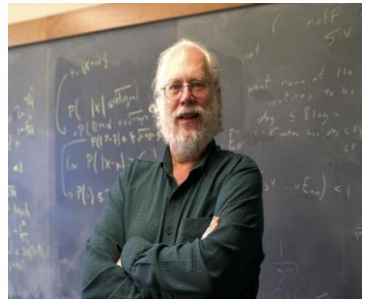
- From traditional (or classical) computers to modern supercomputers, function in the same way. These machines generally perform calculations sequentially, storing data by using binary bits of information. Each bit represents either 0 or 1.
- When combined into binary code and manipulated by using logic operations, we can use computers to create everything from simple operating systems to the most advanced supercomputing calculations.
- By taking advantage of quantum physics, large-scale quantum computers would be able to tackle certain complex problems many times faster than modern classical machines. With a quantum computer, some problems that might take a classical computer thousands of years to solve might be solved in a matter of minutes or hours.
- Quantum mechanics, the study of physics at very small scales, reveals surprising fundamental natural principles. Quantum computers specifically harness these phenomena to access mathematical methods of solving problems not available with classical computing alone.

Evolution of Quantum Computer

- 1982-Richard Feynman-proposed the idea of creating machine based on the laws of quantum mechanics
- 1985-David Deutsch- developed the quantum Turing machine.
- 1994-Peter Shor - came up with a quantum algorithm to factor very large numbers







Breakthrough in Quantum Research

- **Alain Aspect, John F. Clauser, Anton Zeilinger** won Nobel Prize in Physics to pioneering researchers in quantum entanglement "*for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science*

- **Entangled states – from theory to technology**

Alain Aspect, John Clauser and Anton Zeilinger have each conducted groundbreaking experiments using entangled quantum states, where two particles behave like a single unit even when they are separated. Their results have cleared the way for new technology based upon quantum information.

Quantum Computer

- IBM offers the world's most powerful quantum computers via the cloud and powered by Qiskit, the quantum software stack built for performance.
- QpiAI has launched a 25-qubit superconducting quantum computer.
- The system is India’s first full-stack quantum computing platform and forms part of the Bangalore-based startup’s contributions under the country’s National Quantum Mission (NQM).

IBM Quantum Studies

- RIKEN & Cleveland Clinic: Modeling chemistry for the future of drug discovery.
- **E.ON:** Using quantum computing to help tackle electrical grid complexity.
- **JSR:** Charting a new future for the global semiconductor industry.
- **Boeing:** Seeking new ways to engineer strong, lightweight materials.
- **CERN:** Searching for Higgs events and the origins of the universe.
- **Mitsubishi Chemical, JSR, & Keio University:** Exploring new forms of light-emitting materials.
- **Mitsubishi Chemical:** Pursuing game-changing power sources with quantum.
- **Mercedes-Benz:** Envisioning a new generation of electric vehicles through battery technology.

- https://www.ibm.com/quantum/case-studies

How do Quantum Computers work?

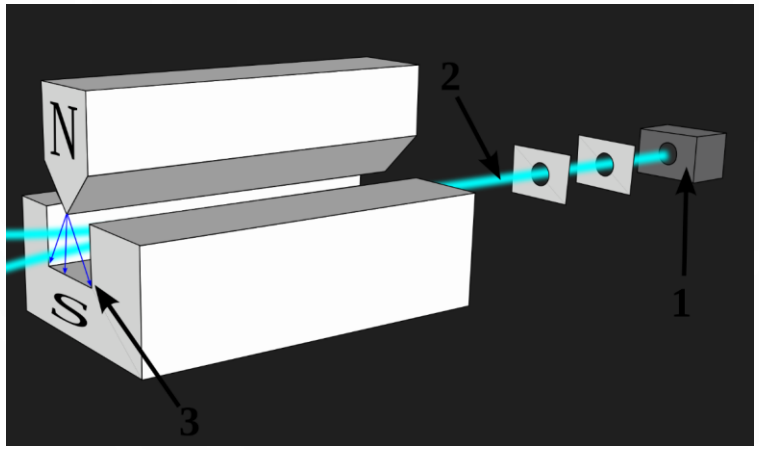
- In traditional or classical computing, a single bit can be thought of as a piece of binary information, notated as either 0 or 1. Modern computers typically represent bits as either an electrical voltage or current pulse.
- In these systems, when there is no current flowing, the circuit can be considered to be off, and this state is represented as 0. When current is flowing, the circuit is considered on, and this state is represented as 1.

Working Principle

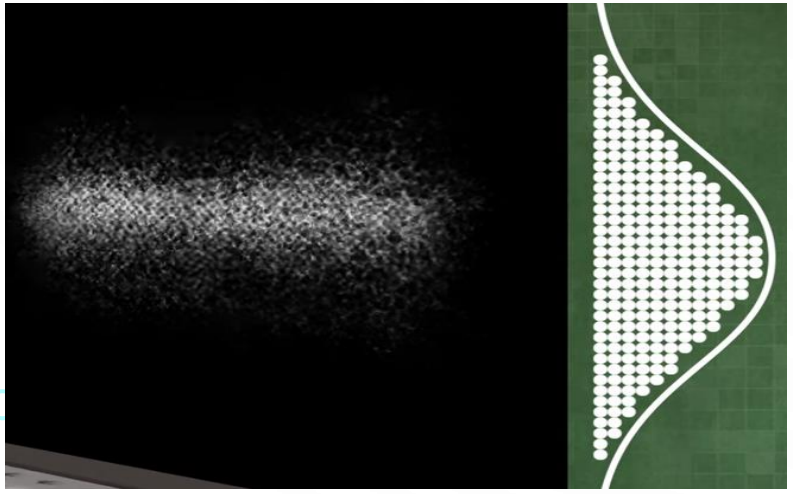
- Imagine you are standing in the center of a complicated maze. To escape the maze, a traditional computer would have to “brute force” the problem, trying every possible combination of paths to find the exit. This kind of computer would use bits to explore new paths and remember which ones are dead ends.
- A Quantum computer **evolves amplitudes** assigned to each possible state. These amplitudes **interfere** constructively for correct answers and destructively for wrong ones. At the end, when **measurement is done,** only **one outcome is obtained.**
- The algorithm is designed to make the correct one highly probable.

Measuring spin (Stern-Gerlach experiment)

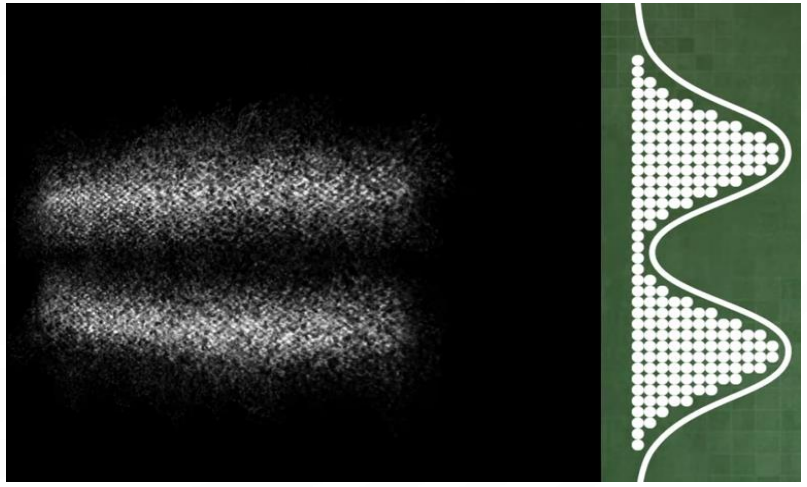
- Experimental Set Up

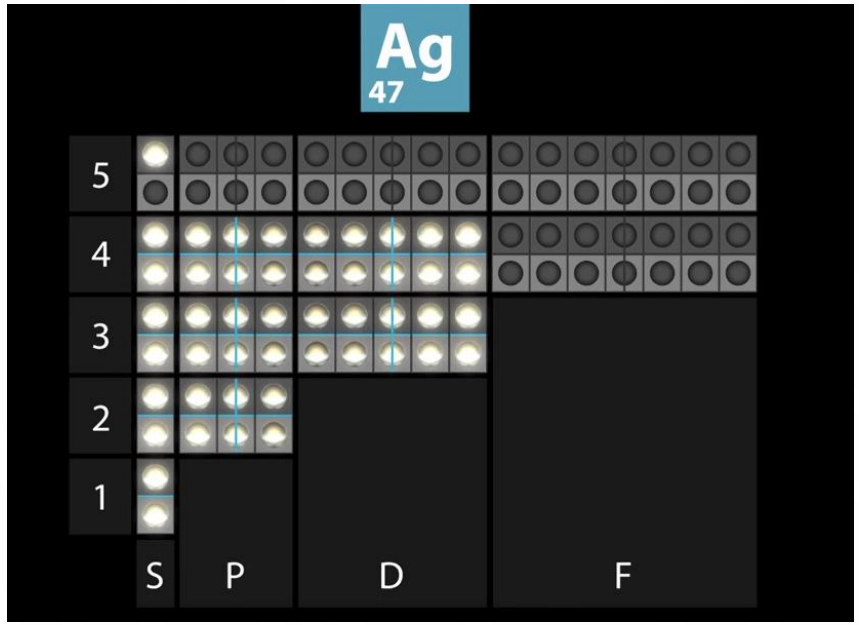


- Classical Outcome



Experimental Observation

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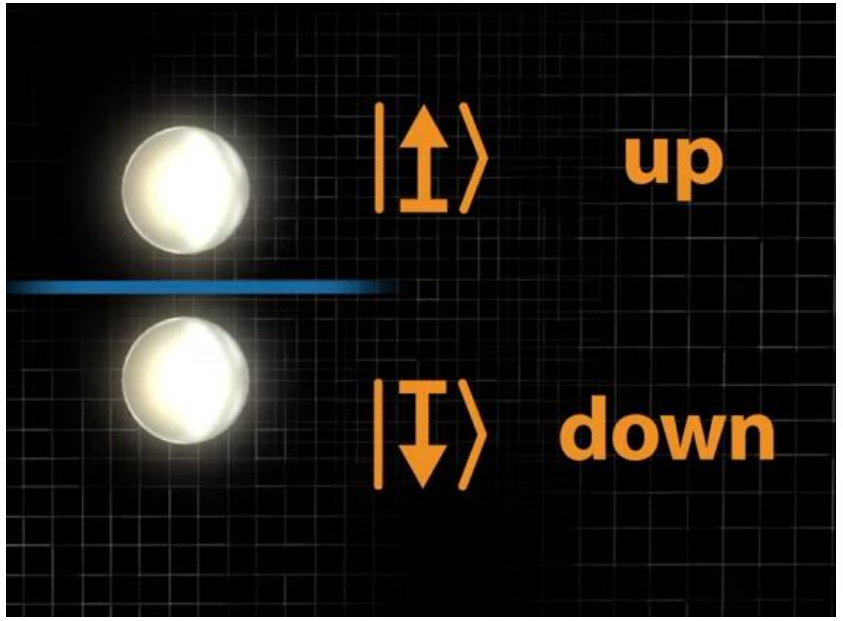
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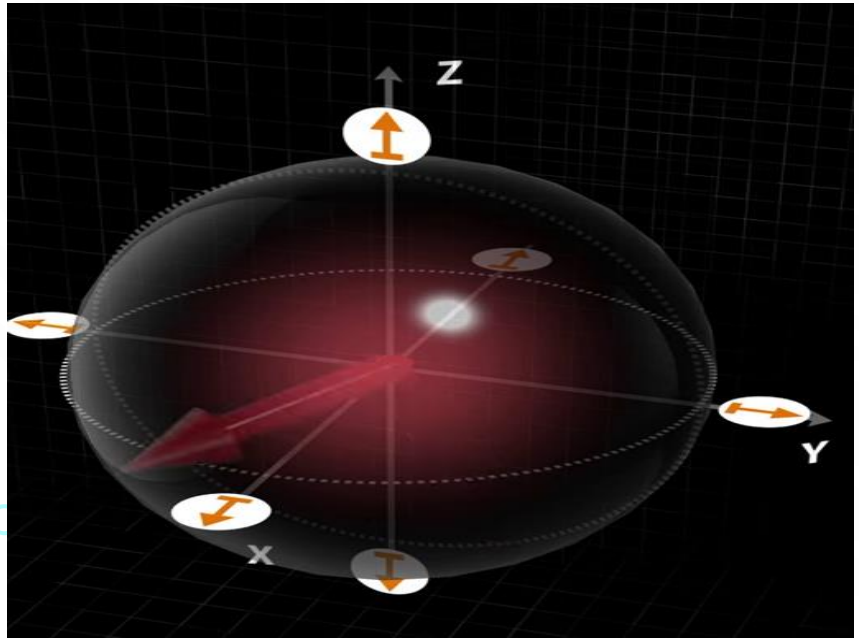
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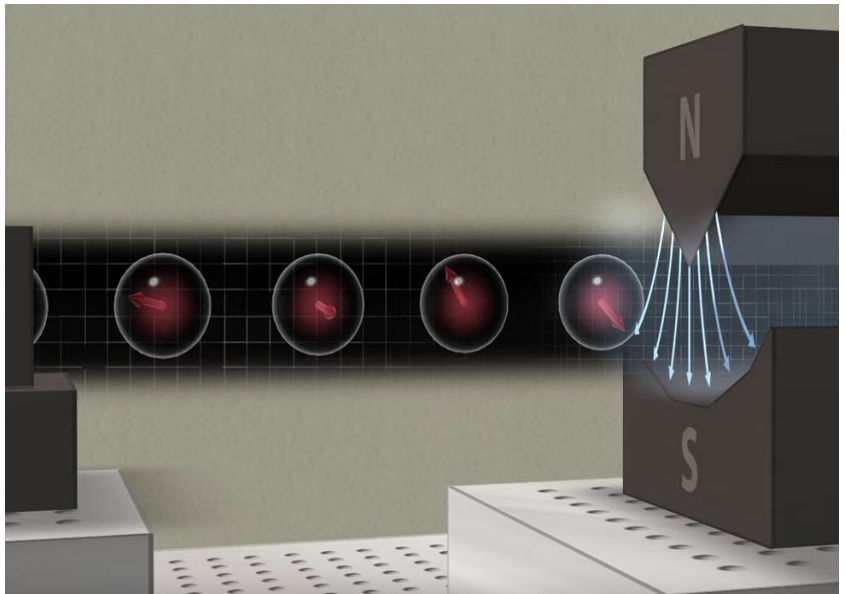
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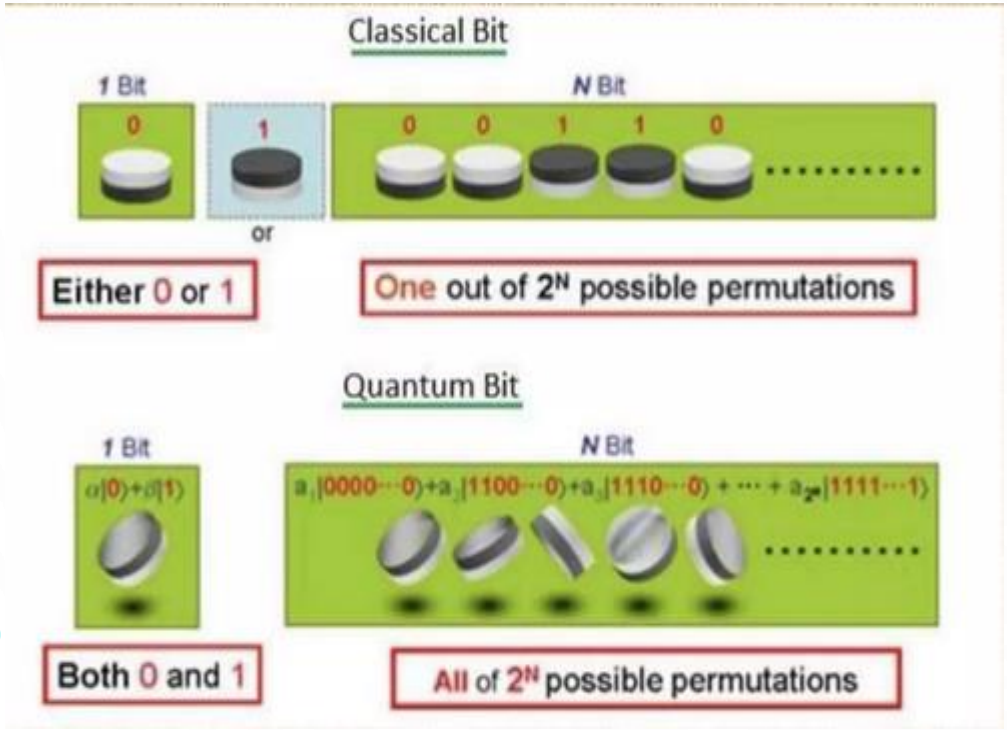
Qubits



- Quantum Computers process data differently by using Quantum Bits (Qubits).
- A Qubit can behave like a bit and store either a zero or a one, but it can also be a weighted combination of zero and one at the same time.
- On the quantum level, Qubit probability is measured as a wave function. The probability amplitude of a Qubit can be used to encode more than one bit of data and carry out extremely complex calculations when combined with other Qubits.

Spin and Qubit

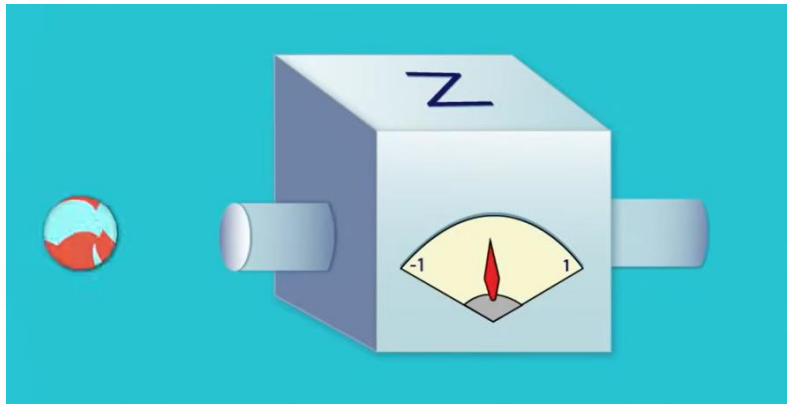
$|0\rangle =$

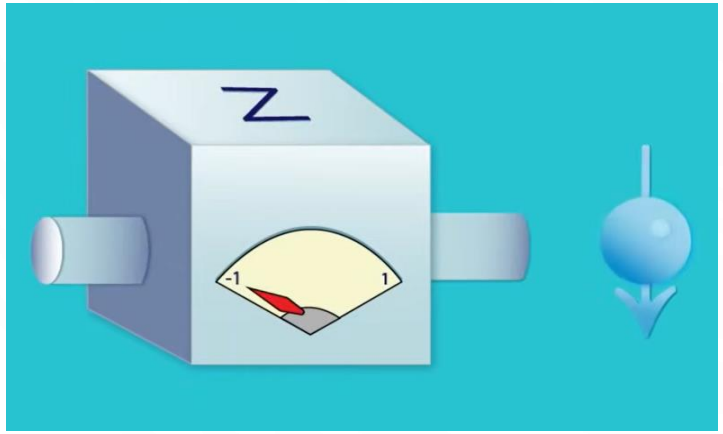
$|1\rangle =$

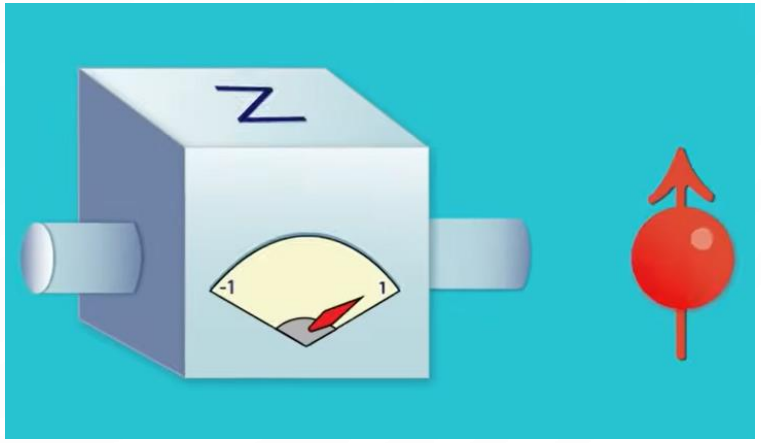
Linear superposition of Qubit

$$\alpha|0\rangle + \beta|1\rangle =$$

Measuring Qubit







Qubits

- A qubit can exist in the state $|0\rangle$ or the state $|1\rangle$ but it can also exist in a *superposition* state.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

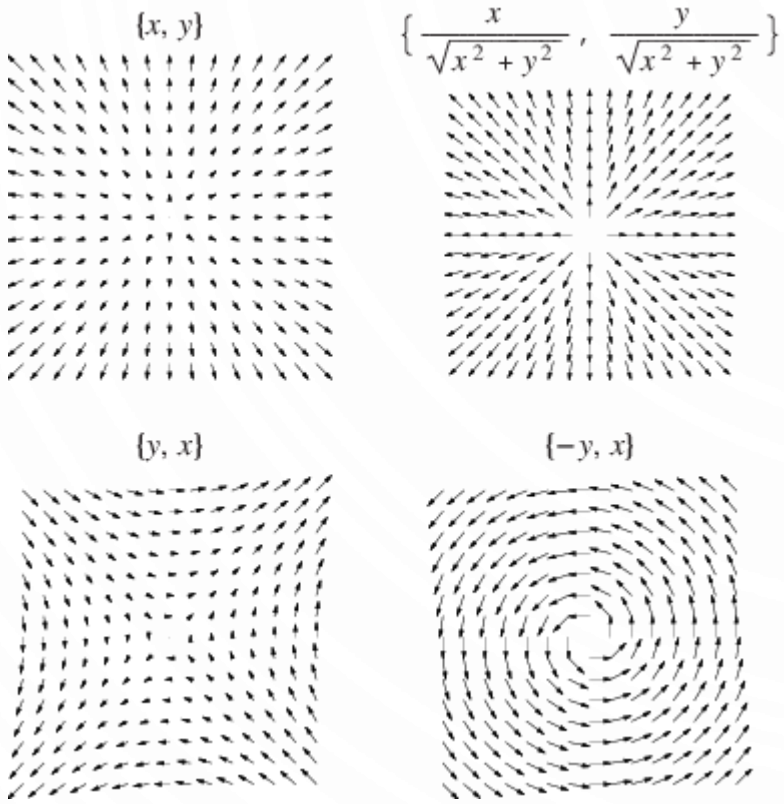
when a qubit is measured, it is only going to be found to be in the state $|0\rangle$ or the state $|1\rangle$

- $|\alpha|^2$: Probability of finding Ψ in state $|0\rangle$
- $|\beta|^2$ Probability of finding Ψ in state $|1\rangle$

$$|\alpha|^2 + |\beta|^2 = 1$$

Linear Algebra

Vector Space



- Arena in which quantum computation takes place is a mathematical abstraction called a vector space
- Quantum states behave mathematically in an analogous way to physical vectors
- A vector space *V* is a nonempty set with elements *u*, *v* called *vectors*.
- Vector addition: An operation that assigns the sum $w = u + v$, which is also an element of *V*, *w* is another vector belonging to the same space

that the vector $\alpha u \in V$

- *Axiom 1:* Associativity of addition. Given vectors u, v, and w,

$$(u + v) + w = u + (v + w)$$

- *Axiom 2:* There is a vector belonging to V called the *zero vector* that satisfies

$$u + 0 = 0 + u = u$$

- *Axiom 3:* For every $u \in V$ there exists an additive inverse of u such that

$$u + (-u) = (-u) + u = 0$$

- *Axiom 4:* Addition of vectors is commutative: $u + v = v + u$

- Quantum computation is the vector space $\mathbb{C}^n$, which is the vector space of "*n*-tuples" of complex numbers.
- *n*-dimensional *column vector*

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

- Vector addition is carried out component by component, producing a new, third vector

$$|a\rangle + |b\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$$

- Multiplication of a vector by a scalar proceeds as

$$\alpha|a\rangle = \alpha \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \vdots \\ \alpha a_n \end{pmatrix}$$

Linear combinations of vectors

- A linear combination of these vectors is given by $\alpha_1|v_1\rangle + \alpha_2|v_2\rangle + \cdots + \alpha_n|v_n\rangle = \sum_{i=1}^{n} \alpha_i|v_i\rangle$

- If a given set of vectors $|v_1\rangle, |v_2\rangle, \ldots, |v_n\rangle$ can be used to represent any vector *u* that belongs to the vector space , we say that the set $\{|v_i\rangle\}$ *spans* the given vector space V

$$|u\rangle = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \gamma \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Linear combinations of vectors

- Therefore we see that the set of vectors defined as

$$|v_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |v_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |v_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

- Spans the space $\mathbb{C}^3$
- If one vector of the set can be written as a linear combination of the other vectors in the set, then the set is *linearly dependent*.

Basis and Dimension

- When a set of vectors is linearly independent and they span the space, the set is known as a basis.
- We can express any vector in the space V in terms of a linear expansion on a basis set. The dimension of a vector space V is equal to the number of elements in the basis set.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\langle b_1 | b_1 \rangle = 1, \langle b_2 | b_2 \rangle = 1, \text{ and } \langle b_1 | b_2 \rangle = 0$$

Basis and Dimension

- Quantum state can be written as a linear combination of a basis set with complex coefficients of expansion as

$$|\psi\rangle = \sum_{i=1}^{n} c_i |v_i\rangle = c_1|v_1\rangle + c_2|v_2\rangle + \cdots + c_n|v_n\rangle$$

- The modulus squared of a given coefficient *ci* gives the probability that measurement finds the system in the state $|v_i\rangle$

Basis and Dimension

- The mathematical model for measuring spin in the vertical direction will be given using the standard basis. Rotating the measuring apparatus will be described mathematically by choosing a new orthonormal basis.
- Direction of the arrow related to the direction of spin.
- “orthonormal” has two parts; ortho from orthogonal, and normal from normalized which, means unit.
- Two-dimensional kets, an orthonormal basis will consist of a set of two unit kets that are orthogonal to one another.
- In general, if we are working with n-dimensional kets, an orthonormal basis consists of a set of n unit kets that are mutually orthogonal to one another.

- There are infinitely many other possible choices

$$\left\{\begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{-1}{\sqrt{2}}\end{bmatrix},\begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}\right\} \text{ and } \left\{\begin{bmatrix}\frac{1}{2}\\ \frac{\sqrt{3}}{2}\end{bmatrix},\begin{bmatrix}\frac{-\sqrt{3}}{2}\\ \frac{1}{2}\end{bmatrix}\right\}$$

$$|\uparrow\rangle = \begin{bmatrix}1\\0\end{bmatrix}, |\downarrow\rangle = \begin{bmatrix}0\\1\end{bmatrix}, |\rightarrow\rangle = \begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{-1}{\sqrt{2}}\end{bmatrix}, |\leftarrow\rangle = \begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}, |\nearrow\rangle = \begin{bmatrix}\frac{1}{2}\\ \frac{-\sqrt{3}}{2}\end{bmatrix},$$

$$\text{and } |\swarrow\rangle = \begin{bmatrix}\frac{\sqrt{3}}{2}\\ \frac{1}{2}\end{bmatrix}.$$

$$\langle\uparrow|\uparrow\rangle = 1 \quad \langle\downarrow|\downarrow\rangle = 1 \quad \langle\uparrow|\downarrow\rangle = 0 \quad \langle\downarrow|\uparrow\rangle = 0$$
$$\langle\rightarrow|\rightarrow\rangle = 1 \quad \langle\leftarrow|\leftarrow\rangle = 1 \quad \langle\rightarrow|\leftarrow\rangle = 0 \quad \langle\leftarrow|\rightarrow\rangle = 0$$
$$\langle\nearrow|\nearrow\rangle = 1 \quad \langle\swarrow|\swarrow\rangle = 1 \quad \langle\nearrow|\swarrow\rangle = 0 \quad \langle\swarrow|\nearrow\rangle = 0$$

Inner Products

- This is a generalization of the dot product used with ordinary vectors in Euclidean space.
- While the dot product takes two vectors and maps them into a real number, in quantum mechanics the inner product will take two vectors from $\mathbb{C}^n$ and map them to a *complex* number.
- If the inner product between two vectors is zero $\langle u|v\rangle = 0$
- $|u\rangle$ and $|v\rangle$ are *orthogonal* to one another.
- The inner product is a complex number.

Uniqueness of a spanning set

- A spanning set of vectors for a given space V is not unique. We can write any arbitrary column vector with two elements in the following way

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Now consider the set $|u_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |u_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ .This set also spans the space

- We can also use these vectors to represent qubits.
- We can write any element of this space in terms of these vectors

$$|\psi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$$

$$= \begin{pmatrix}\alpha\\ 0\end{pmatrix} + \begin{pmatrix}0\\ \beta\end{pmatrix} = \frac{1}{2}\begin{pmatrix}\alpha+\alpha\\ \alpha-\alpha\end{pmatrix} + \frac{1}{2}\begin{pmatrix}\beta-\beta\\ \beta+\beta\end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix}\alpha\\ \alpha\end{pmatrix} + \frac{1}{2}\begin{pmatrix}\alpha\\ -\alpha\end{pmatrix} + \frac{1}{2}\begin{pmatrix}\beta\\ \beta\end{pmatrix} - \frac{1}{2}\begin{pmatrix}\beta\\ -\beta\end{pmatrix}$$

- Now we use the scalar multiplication rule to pull the constants α, β outside of the vectors:

$$\frac{1}{2}\begin{pmatrix}\alpha\\ \alpha\end{pmatrix} + \frac{1}{2}\begin{pmatrix}\alpha\\ -\alpha\end{pmatrix} + \frac{1}{2}\begin{pmatrix}\beta\\ \beta\end{pmatrix} - \frac{1}{2}\begin{pmatrix}\beta\\ -\beta\end{pmatrix}$$

$$= \frac{\alpha}{2}\begin{pmatrix}1\\ 1\end{pmatrix} + \frac{\alpha}{2}\begin{pmatrix}1\\ -1\end{pmatrix} + \frac{\beta}{2}\begin{pmatrix}1\\ 1\end{pmatrix} - \frac{\beta}{2}\begin{pmatrix}1\\ -1\end{pmatrix}$$

$$\left(\frac{\alpha+\beta}{2}\right)\begin{pmatrix}1\\1\end{pmatrix}+\left(\frac{\alpha-\beta}{2}\right)\begin{pmatrix}1\\-1\end{pmatrix}=\left(\frac{\alpha+\beta}{2}\right)|u_1\rangle+\left(\frac{\alpha-\beta}{2}\right)|u_2\rangle$$

Orthonormality

- When the norm of a vector is unity, we say that vector is *normalized*

$$\langle a|a\rangle = 1$$

- If a vector by computing $\langle \tilde{u}|\tilde{u}\rangle = \left(\frac{1}{\sqrt{20}}\langle u|\right)\left(\frac{1}{\sqrt{20}}|u\rangle\right) = \frac{1}{20}\langle u|u\rangle = \frac{20}{20} = 1$ tor by it.

$$|\tilde{u}\rangle = \frac{|u\rangle}{\|u\|} = \frac{1}{\sqrt{20}}|u\rangle$$

$$|\tilde{v}\rangle = \frac{|v\rangle}{\|v\|} = \frac{1}{\sqrt{11}}|v\rangle$$

Orthonormality

- If each element of a set of vectors is normalized and the elements are orthogonal with respect to each other, we say the set is *orthonormal*.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\langle 0|0\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 * 1 + 0 * 0 = 1$$

$$\langle 0|1\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1 * 0 + 0 * 1 = 0$$

$$\langle 1|0\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0 * 1 + 1 * 0 = 0$$

$$\langle 1|1\rangle = \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0 * 0 + 1 * 1 = 1$$

Orthonormality

- Normalized $\langle 0|0\rangle = \langle 1|1\rangle = 1$
- Orthogonal $\langle 0|1\rangle = \langle 1|0\rangle = 0$

Orthonormality

- A vector space can have many basis sets

$$|u_1\rangle = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |u_2\rangle = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\langle u_1|u_2\rangle = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 - 1 = 0$$

- The vectors are not normalized. $\|u_1\| = \|u_2\| = \sqrt{2}.$
- We can create an orthonormal set from these vectors by normalizing them
- Orthonormal basis set for $\mathbb{C}^2$ (and hence for qubits):

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

- Suppose that $\{|u_1\rangle, |u_2\rangle, |u_3\rangle\}$ is an orthonormal basis for a three-dimensional Hilbert space.
- A system is in the state $$|\psi\rangle = \frac{1}{\sqrt{5}}|u_1\rangle - i\sqrt{\frac{7}{15}}|u_2\rangle + \frac{1}{\sqrt{3}}|u_3\rangle$$
- (a) Is this state normalized?
- (b) If a measurement is made, find the probability of finding the system in each of the states $\{|u_1\rangle, |u_2\rangle, |u_3\rangle\}$

Postulates

- Postulate 1 (The State of a Quantum System): The state of a quantum system is a unit vector (the state vector) in a complex inner product space (Hilbert space, more generally).
- Postulate 2 (Observables of a Quantum System): The observables of a quantum system (such as position, momentum, spin components) are described by (Hermitian operators) on the quantum state space.

Matrices and Operators

- An *operator* is a mathematical rule that can be applied to a function to transform it into another function. Operators are often indicated by placing a caret above the symbol used to denote the operator. For example, we can define the derivative operator as

$$\hat{D} = \frac{d}{dx}$$

$$\hat{D}f = \hat{D}(x\cos x) = \frac{d}{dx}(x\cos x) = \cos x - x\sin x$$

- This idea can be extended to vector spaces $\hat{A}|\psi\rangle = |\phi\rangle$

- We say that an operator is *linear* if the following relationship holds given complex numbers α and β and state vectors

$$\hat{A}(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) = \alpha(\hat{A}|\psi_1\rangle) + \beta(\hat{A}|\psi_2\rangle)$$

Observables

- In quantum theory, dynamical variables like position, momentum, angular momentum and energy are called *observables*.
- It turns out that an important postulate of quantum theory is that there is an operator that corresponds to each physical observable.
- **The Pauli Operators**
- Operators can act on qubits, which as we have seen are just kets or state vectors in a two-dimensional vector space. A set of operators that turns out to be of fundamental importance in quantum computation is known as *Pauli operators*.

Spin observables of a spin-1/2 system-I

- Starting with the first of the Pauli operators, we have the identity operator.

$$\sigma_0|0\rangle = |0\rangle, \quad \sigma_0|1\rangle = |1\rangle$$

$$\sigma_1|0\rangle = |1\rangle, \quad \sigma_1|1\rangle = |0\rangle$$

- The Pauli Matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Spin observables of a spin-1/2 system-II

- Basis states are given by

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- As noted above, since X transforms one basis state into the other, it is sometimes called the NOT operator or *bit flip* operator.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Spin observables of a spin-1/2 system-III

- The operator Sz represented by the matrix Sz = $\frac{\hbar}{2}$ Z
- Note that the action of Sz on $|0\rangle$ or $|1\rangle$ is simply a scalar + $\frac{\hbar}{2}$, - $\frac{\hbar}{2}$ times $|0\rangle$ or $|1\rangle$
- The spin-up and spin-down states corresponding to the Z-axis are eigenstates of the operator Sz.
- Sz represents the physical observable corresponding to the Z-component of the spin of the spin-1/2 system

Spin observables of a spin-1/2 system-IV

- S_z represents the physical observable corresponding to the Z-component of the spin of the spin-1/2 system.
- Similarly, we can represent the observables corresponding to the X-component and Y -component of the spin.

Spin observables of a spin-1/2 system-V

$$Y|0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0*1 - i*0 \\ i*1 + 0*0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle$$

$$Y|1\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0*0 - i*1 \\ i*0 + 0*1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i|0\rangle$$

$$Z|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1*1 + 0*0 \\ 0*1 - 1*0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

$$Z|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1*0 + 0*1 \\ 0*0 - 1*1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -|1\rangle$$

Hermitian operator

- An operator $\hat{A}$ is said to be *Hermitian* if $\hat{A} = \hat{A}^{\dagger}$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{\dagger} = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

Hermitian, Unitary operators

- Note that the matrices X, Y and Z are Hermitian and unitary.
- The operator Sz is Hermitian. It is equal to its conjugate-transpose.

$$(S_z)^T)^* = S_z$$

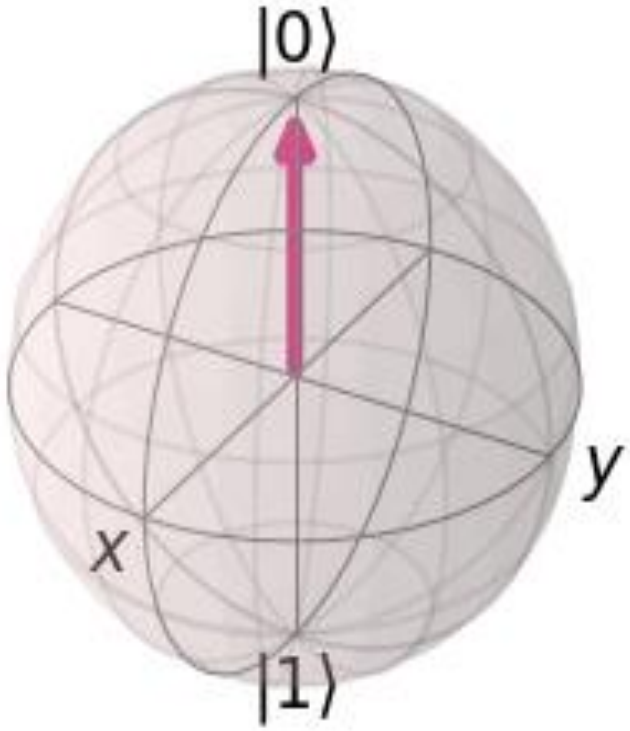
- Verify that Sx is also a Hermitian matrix.

Hermitian operators

- Hermitian operators have real eigenvalues. These eigenvalues represent the physical values of the corresponding observables.
- For a spin-1/2 system, the spin along the Z-axis can take one of two values $\pm\hbar/2$ corresponding to the states $|0\rangle$ and $|1\rangle$ respectively.
- The spin along the X-axis can take one of two values $\pm\hbar/2$ corresponding to the states $|+\rangle$ and $|-\rangle$ respectively.

Qubit representation

BLOCH SPHERE



Single qubit: $|q\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix}\alpha\\ \beta\end{bmatrix} \qquad \alpha, \beta \in \mathbb{C}$

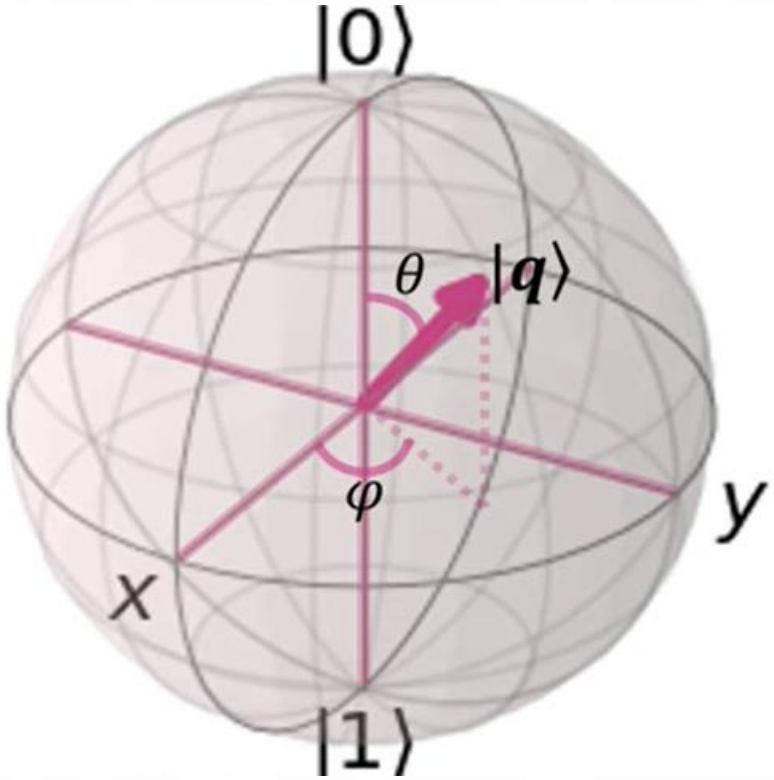
$$\alpha = r_1 e^{i\theta_1} \qquad \beta = r_2 e^{i\theta_2}$$

$$r_1{}^2 + r_2{}^2 = 1$$

$$\theta_2 = \theta_1 + \varphi$$

$$|q\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$

BLOCH SPHERE



$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix} \qquad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{bmatrix}$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix}$$

$$|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = \begin{bmatrix}1/\sqrt{2}\\i/\sqrt{2}\end{bmatrix}$$

$$|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) = \begin{bmatrix}1/\sqrt{2}\\-i/\sqrt{2}\end{bmatrix}$$

Complex number

- Standard form

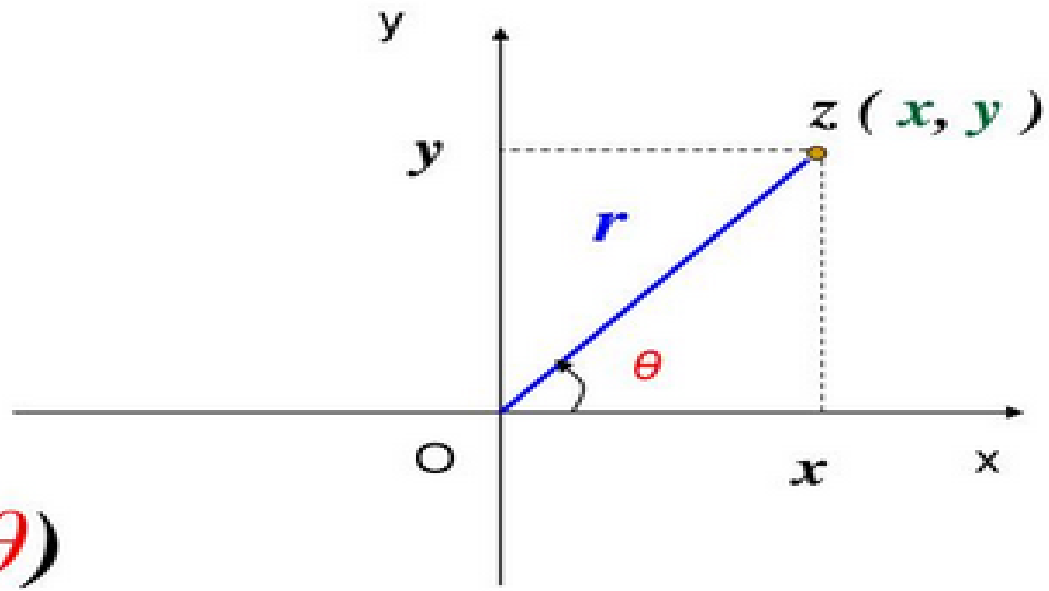
$$z = x + y\,i$$

- Polar form

$$z = r\,(\cos\theta + i\,\sin\theta)$$

- Exponential form

$$z = re^{\,i\,\theta}$$



$$r = \sqrt{x^2 + y^2}$$

$$Arg z = \arctan\left(\frac{y}{x}\right) \in (-\pi, \pi]$$

Argand Diagram

TENSOR PRODUCTS

- In quantum mechanics we don’t always work with single particles in isolation. In many cases, some of which are seen in the context of quantum information processing, it is necessary to work with multiparticle states. Mathematically, to understand multiparticle systems in quantum mechanics, it is necessary to be able to construct a Hilbert space H that is a composite of the independent Hilbert spaces that are associated with each individual particle. The machinery required to do this goes by the name of the *Kronecker* or *tensor product*

- In this section we describe how to calculate tensor products when state vectors are written in a matrix representation, meaning as column vectors. We are primarily concerned with going from $\mathbb{C}^2 \to \mathbb{C}^4$

$$|\phi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{and} \quad |\chi\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$|\phi\rangle \otimes |\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}$$

- Note that the order of the tensor product is not relevant

- Calculate the tensor product of

$$|a\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \quad \text{and} \quad |b\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix}\sqrt{2}\\1\end{pmatrix}$$

TENSOR PRODUCTS OF MATRICES

- We focus on taking tensor products of operators in two-dimensional Hilbert spaces to produce an operator that acts on a four-dimensional Hilbert space.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}$$

- Find the tensor product of the Pauli matrices X and Z

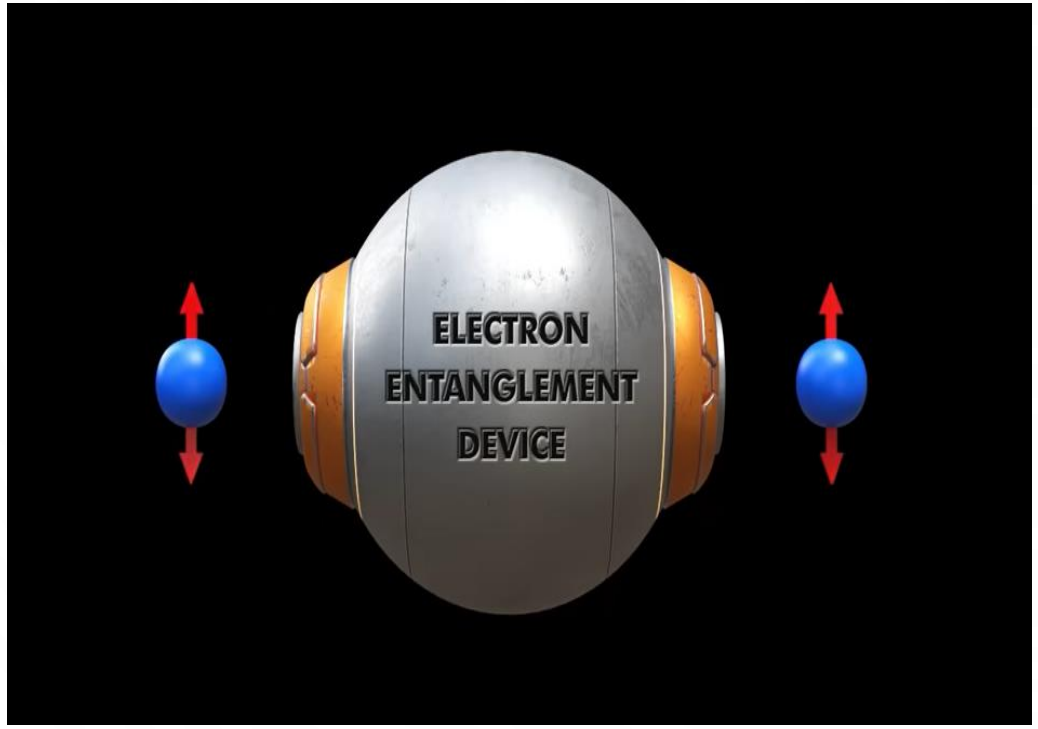
$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

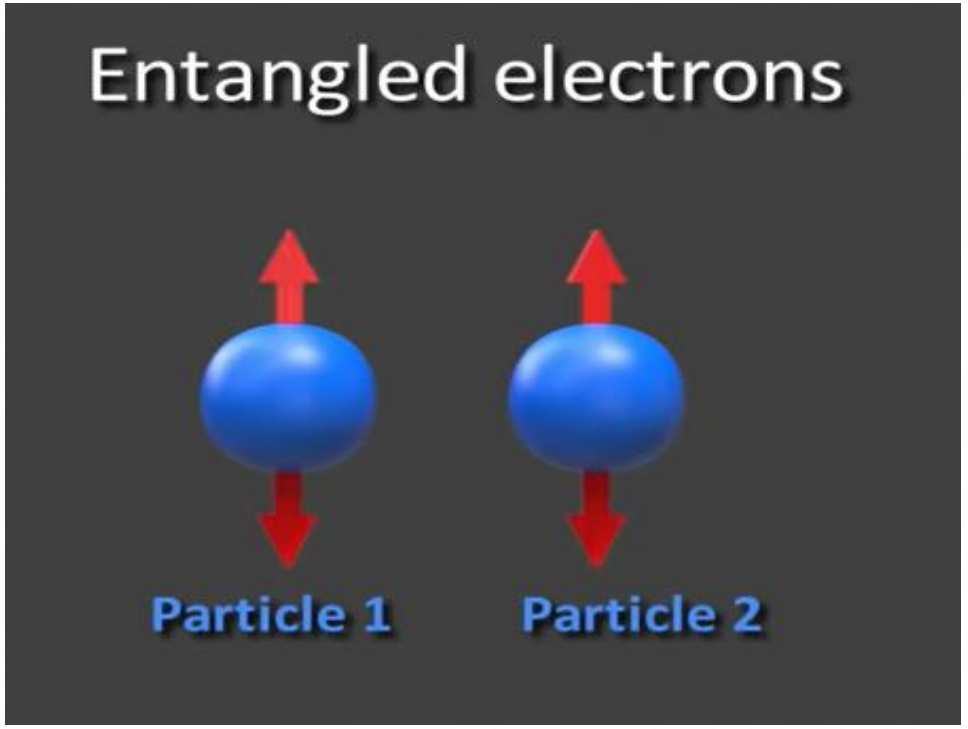
$$X \otimes Z = \begin{pmatrix} (0)Z & (1)Z \\ (1)Z & (0)Z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Entanglement-I

- One of the most unusual and fascinating aspects of quantum mechanics is the fact that particles or systems can become *entangled*.
- For the simplest two quantum systems case we denote the systems A and B. If these systems are entangled, this means that the values of certain properties of system A are correlated with the values that those properties will assume for system B. The properties can become correlated even when the two systems are spatially separated—leading to the phrase *spooky action* at a distance.

Entanglement-II



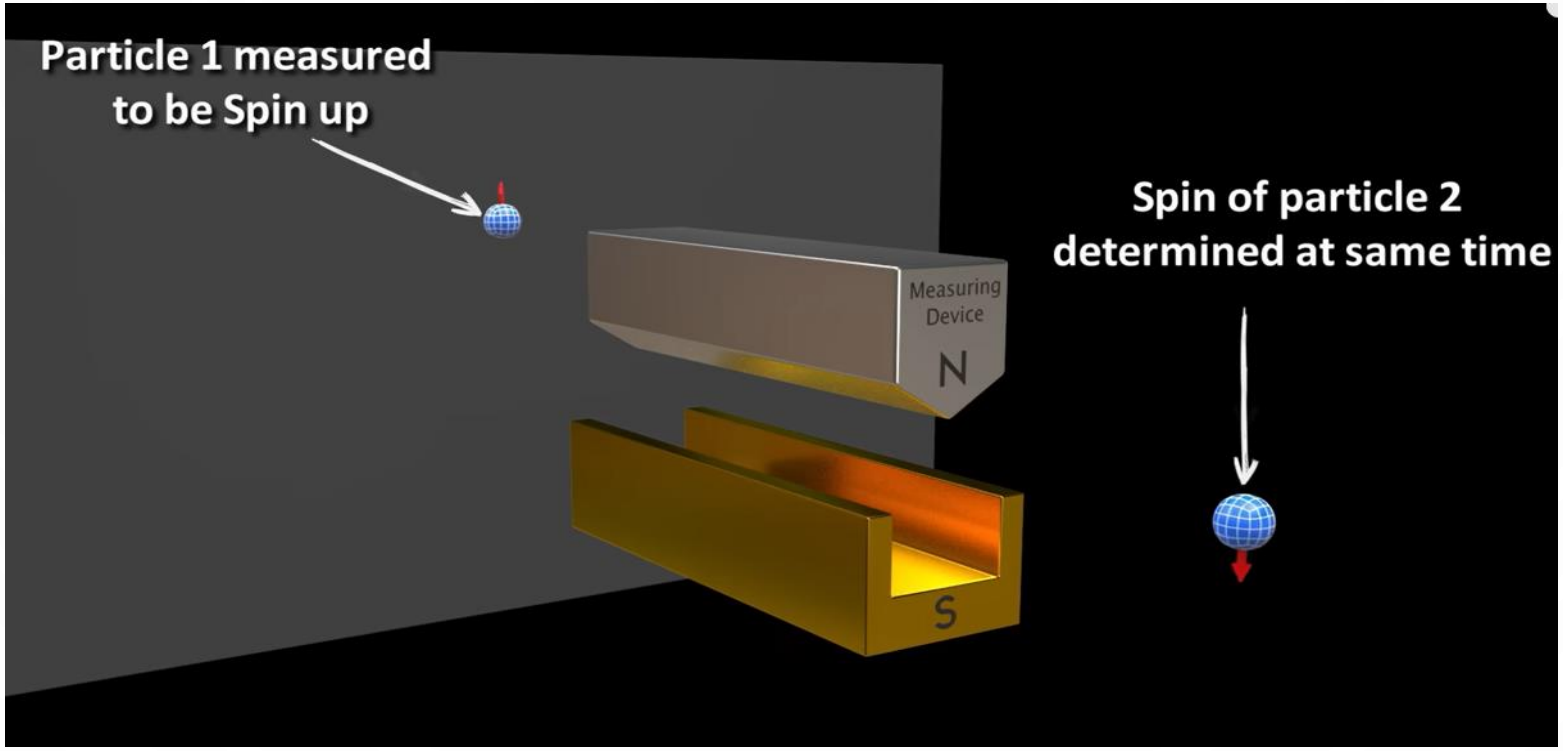


Ref. Online lecture video

Entanglement-III

- A neutral Pi meson sometimes decays into a positron and an electron.
- Observe the deflection of the outcoming particles under the influence of an inhomogeneous magnetic field. An inhomogeneous field used to deflect spins is often called a Stern-Gerlach device, to gather evidence of the existence of quantum mechanical spin.
- As the meson decays, an electron is kicked out in one direction, and a positron in the other.
- If we have a source of many mesons, we can collect statistics on this.

Entanglement-IV



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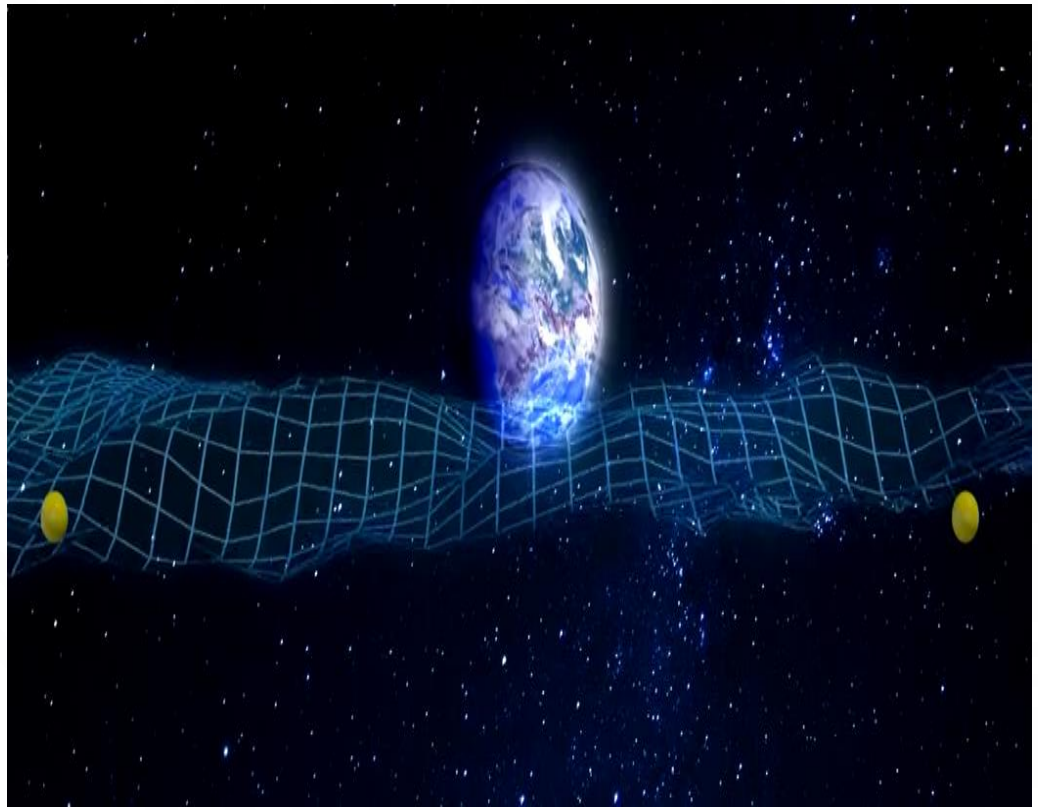
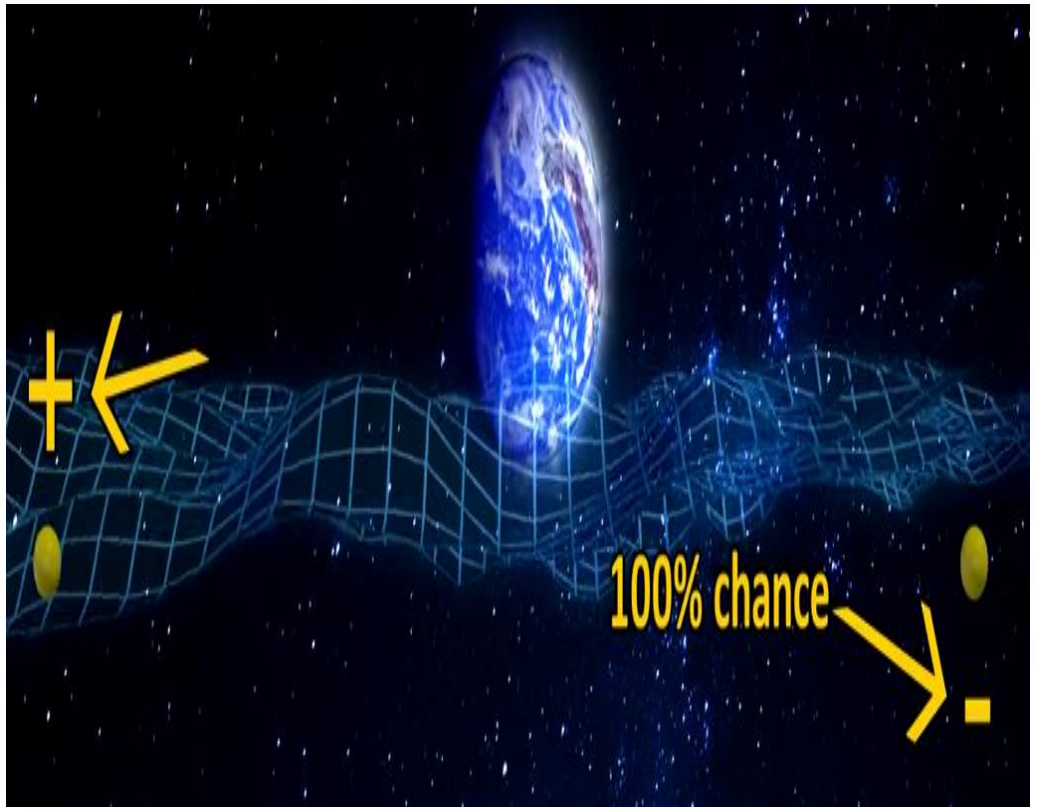
Entanglement-V





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EPR

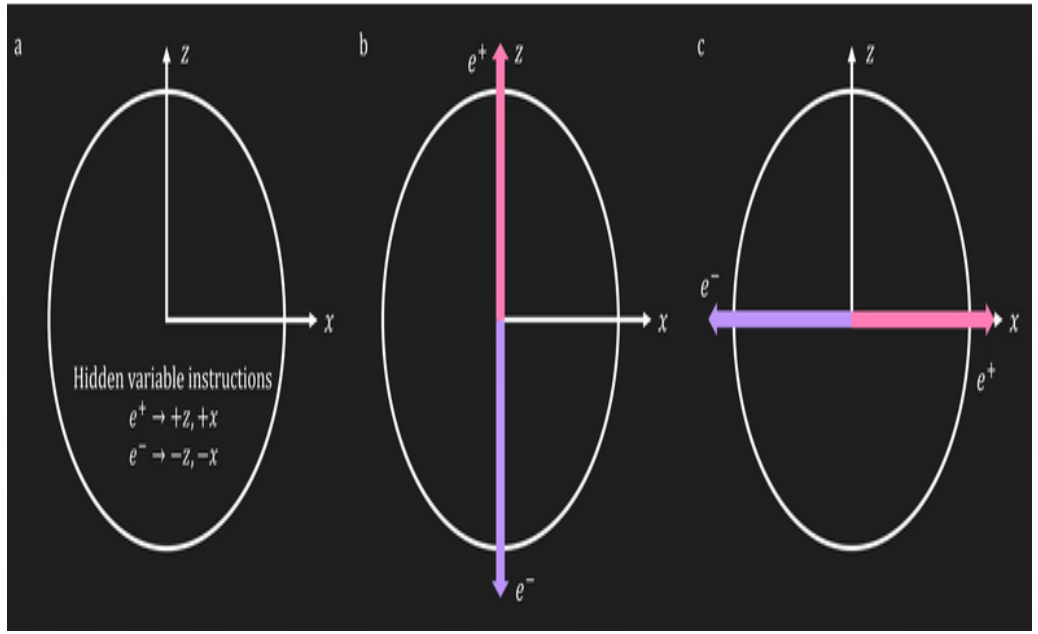
- Einstein and two colleagues, Podolsky and Rosen (now commonly known as EPR), published a paper titled “Can the quantum-mechanical description of reality be considered complete?” This paper—written by the quantum skeptic Einstein—was actually designed to show that quantum theory is incomplete and to make absurd predictions.
- A core value held by EPR and other “realists” was that the properties of physical systems have definite values (an *objective reality*) whether you observe the system or not. Another way to say this is that a given property of a system has a sharply defined value before a measurement is made.

EPR

Ref. Online lecture video

- Quantum mechanics, however, tells a different story. Quantum theory tells us that the wavefunctions of each particle exist in superpositions and each property does not have a definite value until we measure it. Since quantum theory cannot tell one the definite values of each property prior to measurement, EPR said the theory was *incomplete*. There could be some other physical variables we don’t yet know about that would allow us to describe the definite properties of each particle using the theory. Because we don’t know what those variables are, they are called *hidden variables*.

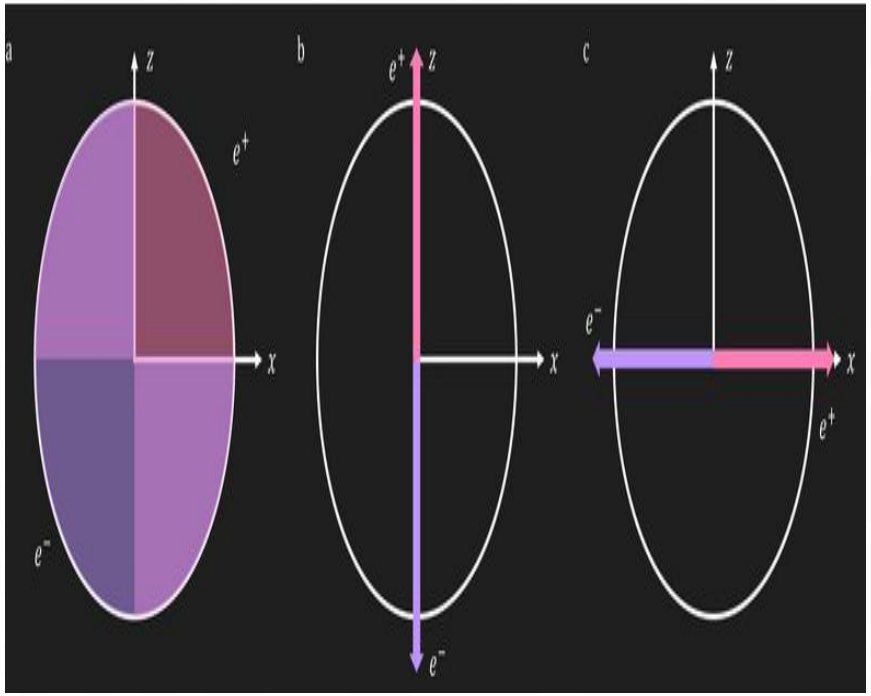
Hidden Variable



- The weakest statement of Einstein's option is that there is some set of instructions that are stored in the spin that determine what the measurements outcomes will be when measured along any axis.

https://quantum.cloud.ibm.com/learning/en/modules/quantum-mechanics/bells-inequality-with-qiskit

Quantum Mechanics



- The directions of the positron and electron spins are smeared out in a distribution of probability and have no definite direction. Each spin could actually have a non-zero probability of pointing in any direction as long as they are opposite each other. Note that in the case of Born's option, it is still true that angular momentum must be conserved. So if one wave of probability is "collapsed" such that the spin points along +z, the other will point along −z and be deflected in the opposite direction.

https://quantum.cloud.ibm.com/learning/en/modules/quantum-mechanics/bells-inequality-with-qiskit

BELL'S Theorem

- In 1964 these ideas were taken much further by the brilliant theoretical physicist John S. Bell. By considering spin measurements along three nonorthogonal directions, which are conventionally labeled **a, b,** and **c,** and derived an inequality that is satisfied by local realistic theories but violated by quantum mechanics
- An ensemble or large number of systems have been prepared so that Alice and Bob can measure spin along the three directions **a, b,** and **c**. With three directions to consider and 2 possible states.

Quantum Computing Explained: David McMahon

TABLE 7.2 Measurements for particles Alice and Bob

	Alice			Bob		
Population	a	b	c	a	b	c
N_1	+	+	+	−	−	−
N_2	+	+	−	−	−	+
N_3	+	−	+	−	+	−
N_4	+	−	−	−	+	+
N_5	−	+	+	+	−	−
N_6	−	−	+	+	+	−
N_7	−	+	−	+	−	+
N_8	−	−	−	+	+	+

Quantum Computing Explained: David McMahon

- The total number of particles is denoted by *N*, where there are *Ni* particles found in state *I*
- $$N = N_1 + N_2 + N_3 + N_4 + N_5 + N_6 + N_7 + N_8$$
- The direction along which Alice and Bob choose to measure is completely random. But, when deriving Bell's inequality, we assume that Alice and Bob measure along different directions.
- How many cases occur when Alice measures+1 along **a** and Bob measures−1 along **b**- N_1 and N_2
- If Bob chooses to measure along **c,** then he obtains a−1 for populations N_1 and N_3.

- The probability of the two observers measuring the same sign for rows 1 and 8 are zero. For rows 2-7, there are 4 ways to obtain the same sign, which we will show only for row 2.
- Same signs: ac, bc, ca, cb, Opposite signs: aa, ab, ba, bb, cc

$$x \geq 0, y \geq 0, z \geq 0 \qquad x + y \leq x + y + z$$

$$N_3 + N_4 \leq (N_3 + N_4) + (N_2 + N_7) = (N_2 + N_4) + (N_3 + N_7)$$

$$\frac{N_3 + N_4}{N} \leq \frac{(N_2 + N_4)}{N} + \frac{(N_3 + N_7)}{N}$$

- We see that *N3* and *N4* both have Alice measuring +1 along **a** and Bob measuring+1 along **b**

$$\frac{N_3 + N_4}{N} = \Pr(+\mathbf{a}; +\mathbf{b})$$

- That is, this is the probability that Alice measures+1 along **a** *and* Bob measures +1 along **b**.
- Look at N_2 and N_4. Again, we see that Alice measures+1 along **a**. This time Bob gets+1 along **c** in both cases

$$\frac{N_2 + N_4}{N} = \Pr(+\mathbf{a}; +\mathbf{c})$$

- Consider N_3 and N_7. Alice measures a+1 when measuring along **c** in both cases, while Bob obtains a +1 when measuring along **b** in both cases

$$\frac{N_3 + N_7}{N} = \Pr(+\mathbf{c}; +\mathbf{b})$$

$$\Pr(+\mathbf{a};\, +\mathbf{b}) \leq \Pr(+\mathbf{a};\, +\mathbf{c}) + \Pr(+\mathbf{c};\, +\mathbf{b})$$

- Local realistic theories *satisfy* or obey Bell's inequality. quantum mechanics *does not.*
- To see what quantum mechanics says about the situation, we need to consider

a qubit oriented in an arbitrary direction. We prepare the system in the singlet state.

What does Born's option (non-deterministic quantum mechanics) predict?

- We consider the initial state after the decay of the meson and before any measurement as along direction a

$$|\psi\rangle = \frac{|+_a\rangle|-_a\rangle - |-_a\rangle|+_a\rangle}{\sqrt{2}}$$

$$\Pr(+\mathbf{a};+\mathbf{c}) = \left|\frac{\langle +_c|-_a\rangle}{\sqrt{2}}\right|^2 = \frac{1}{2}\sin^2\left(\frac{\theta_{ac}}{2}\right)$$

$$\Pr(+\mathbf{a};+\mathbf{b}) = \frac{1}{2}\sin^2\left(\frac{\theta_{ab}}{2}\right), \quad \Pr(+\mathbf{c};+\mathbf{b}) = \frac{1}{2}\sin^2\left(\frac{\theta_{cb}}{2}\right)$$

- We then obtain

$$\sin^2\left(\frac{\theta_{ab}}{2}\right) \leq \sin^2\left(\frac{\theta_{ac}}{2}\right) + \sin^2\left(\frac{\theta_{cb}}{2}\right)$$

$$\theta_{ac} = \theta_{cb} = \theta,\ \theta_{ab} = 2\theta$$

- Bell's inequality is violated when $0 < \theta < \frac{\pi}{2}$
- Suppose that Alice and Bob design their system such that $\theta = \pi/3$. Bell's inequality is then the nonsensical statement that $0.75 \leq 0.5$

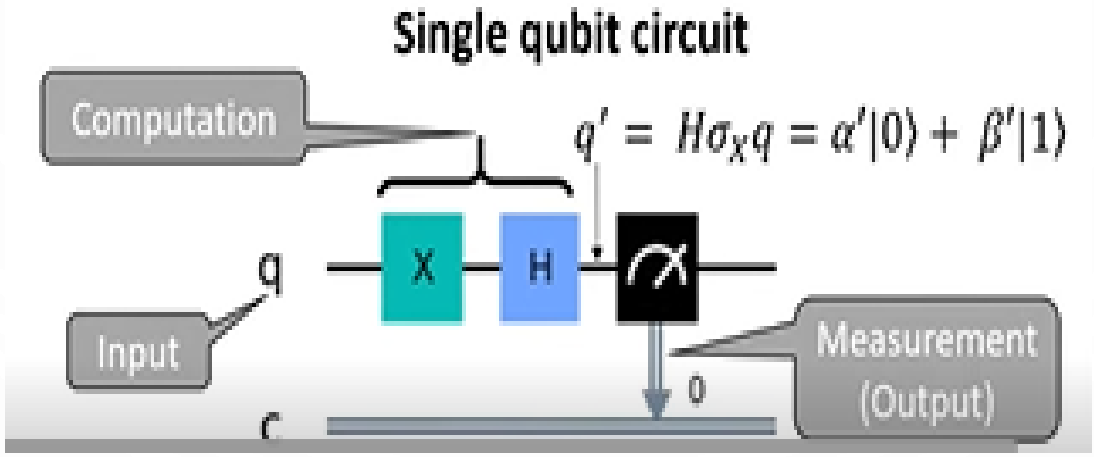
CONTROLLED HADAMARD GATE

$$CH|01\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle$$

$$CH|11\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{pmatrix}$$

$$CH = |00\rangle\langle 00| + |01\rangle\langle 01| + \frac{1}{\sqrt{2}}(|10\rangle\langle 10| + |10\rangle\langle 11| + |11\rangle\langle 10| - |11\rangle\langle 11|)$$

Example of a typical quantum circuit

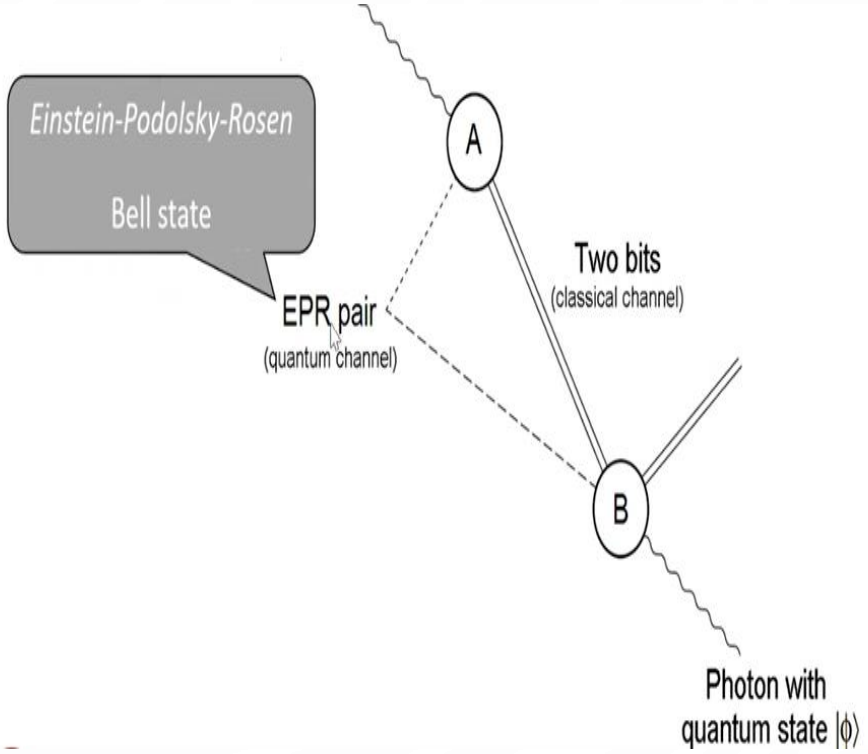


Applications of entanglement: teleportation

- Entanglement to accomplish communications and information processing tasks that would otherwise not be possible.
- *Teleportation*, is a procedure that allows one party (Alice) to send a quantum state to her friend (Bob) without that state being transmitted in the usual sense.
- By using entanglement, Alice and Bob can set up a quantum communications channel that links them together in a quantum way via the EPR paradox—allowing Alice to send her state to Bob in an almost magical fashion.
- Faster than light communication isn’t possible using teleportation because it is still necessary for Alice and Bob to maintain a classical communications link in order for teleportation to work.

Teleportation

- Chinese scientists have sent a packet of information from Tibet to a satellite in orbit, up to 870 miles (1,400 kilometers) above the Earth's surface.
- The scientists beamed the quantum state of a photon (information about how it is polarized) into orbit.



Bell States: four 2-qubit states that are maximally entangled

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\psi_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|\psi_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

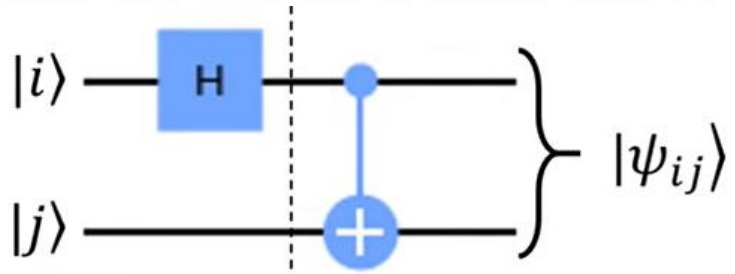
Measurement collapses state to $|00\rangle$ or $|11\rangle$

If we measure the first qubit, we automatically know the state of the second

High amount of correlation, regardless of distance: *useful in quantum computations*

Entanglement: circuit to BELL states

Input $\lvert ij\rangle$	Output $\lvert\psi_{ij}\rangle$
$\lvert 00\rangle$	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle)$
$\lvert 01\rangle$	$\frac{1}{\sqrt{2}}(\lvert 01\rangle + \lvert 10\rangle)$
$\lvert 10\rangle$	$\frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 11\rangle)$
$\lvert 11\rangle$	$\frac{1}{\sqrt{2}}(\lvert 01\rangle - \lvert 10\rangle)$



$|00\rangle$
$|i\rangle = |0\rangle$
$|j\rangle = |0\rangle$
$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$
$= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$
$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|10\rangle$
$|i\rangle = |1\rangle$
$|j\rangle = |0\rangle$
$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|0\rangle$
$= \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$
$= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

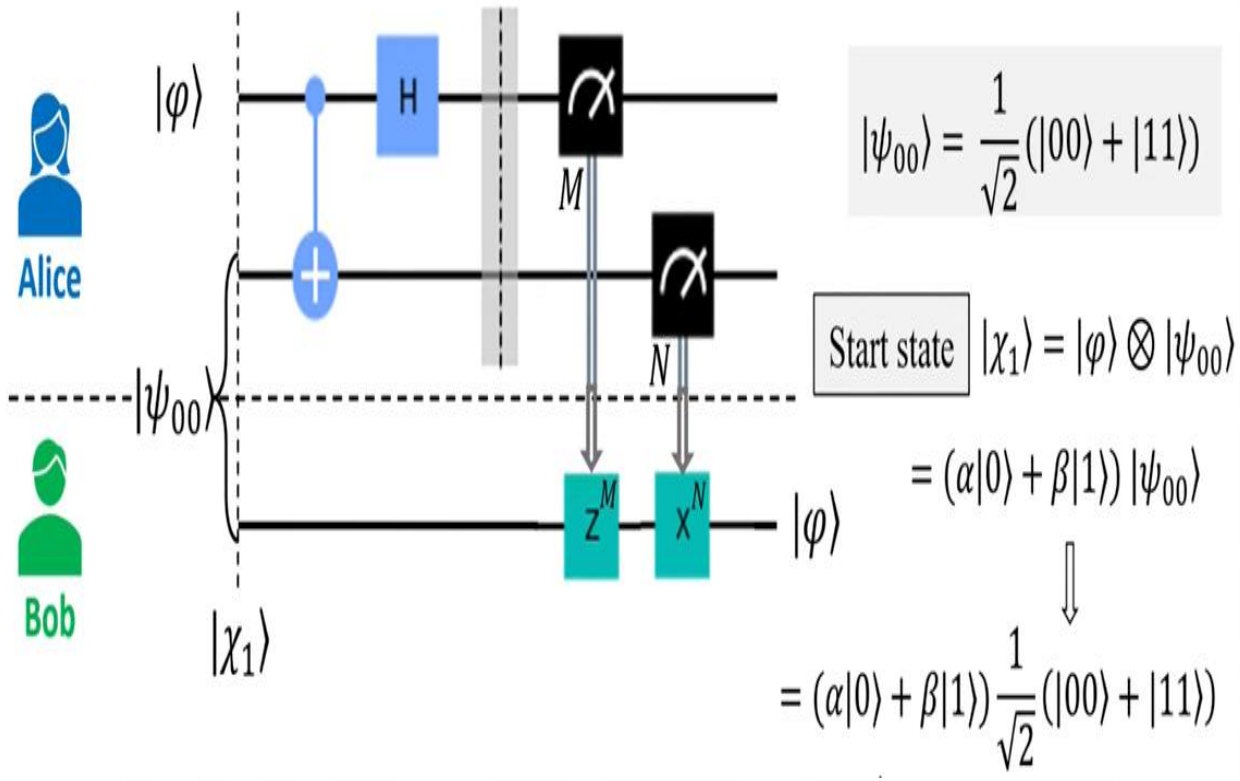
Teleportation step 1: Alice and Bob share an entangled pair of particles

- Alice and Bob create the entangled state

$$|\beta_{00}\rangle = \frac{|0_A\rangle|0_B\rangle + |1_A\rangle|1_B\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

- First member of the pair belongs to Alice and the second member of the pair belongs to Bob. Now Alice and Bob physically separate. Alice decides that she wants to send the state $|\chi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob.

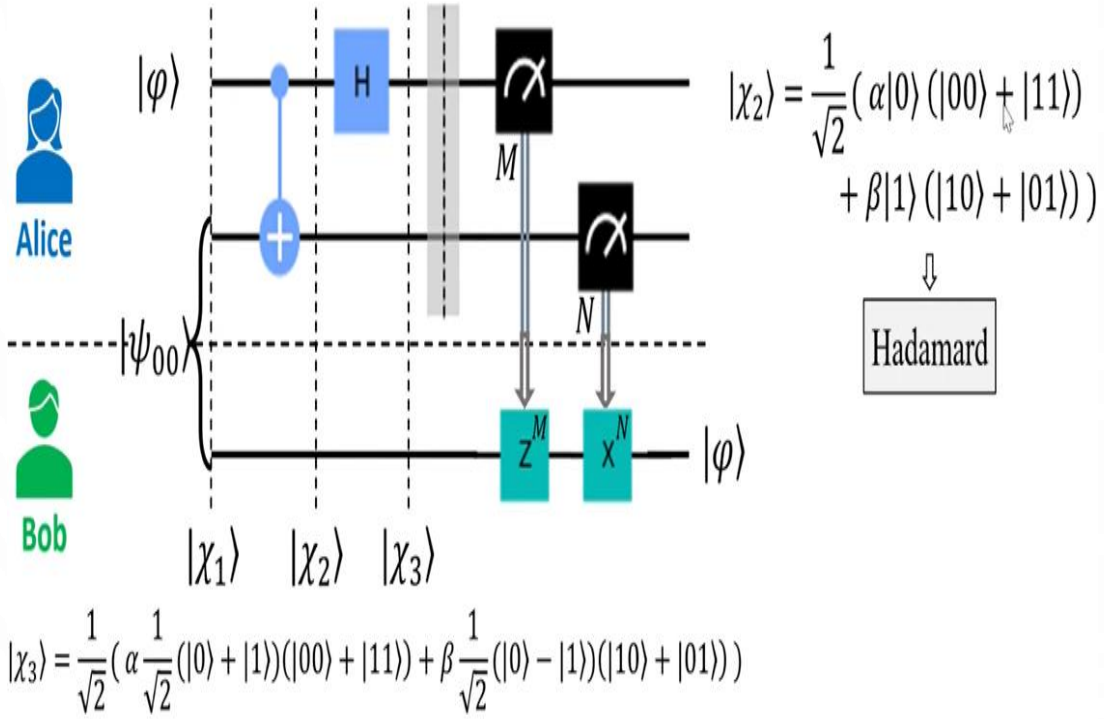
TELEPORTATION STEP 2:



- The first two qubits in this state belong to Alice, while the third rightmost qubit belongs to Bob. So |011> indicates that Alice has a 01 in her possession while Bob has a 1.
- Alice begins interacting her member of the EPR pair, which is the second qubit with the unknown state—the first qubit by applying a CNOT gate.
- She uses the unknown state χ as the control qubit and her member of the EPR pair as the target qubit.
- If the control qubit is 0, nothing happens; if the control qubit is 1, the target qubit is flipped

$$\frac{1}{\sqrt{2}}(\,\alpha|0\rangle\,(|00\rangle + |11\rangle) + \beta|1\rangle\,(|10\rangle + |01\rangle)\,)$$

Teleportation step 3: Alice applies a Hadamard gate



- Next Alice will apply a Hadamard gate to the first qubit. Remember, a Hadamard gate acts on the computational basis states by turning them into superpositions.
- Remember, Bob is in possession of the third qubit.

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\begin{aligned} = \frac{1}{2}(& |00\rangle\,(\alpha|0\rangle + \beta|1\rangle) \\ & + |01\rangle\,(\alpha|1\rangle + \beta|0\rangle) \\ & + |10\rangle\,(\alpha|0\rangle - \beta|1\rangle) \\ & + |11\rangle\,(\alpha|1\rangle - \beta|0\rangle)\ ) \end{aligned}$$

Teleportation step 4: Alice measures her pair

- Alice makes a measurement on both qubits in her possession.
- The state so that the state is written in terms of the possible measurement results on the first two qubits.
- These possible measurement results are

Alice's Measurement MN	Bob's State $\|\varphi'\rangle$
00	$\alpha\|0\rangle + \beta\|1\rangle$
01	$\alpha\|1\rangle + \beta\|0\rangle$
10	$\alpha\|0\rangle - \beta\|1\rangle$
11	$\alpha\|1\rangle - \beta\|0\rangle$

- If Alice measures $|00\rangle$, then the state collapses and Bob has the unknown state $|\chi\rangle = \alpha|0\rangle + \beta|1\rangle$ that Alice wanted to send Bob in the first place in his possession.
- Next, if Alice measures $|01\rangle$, then Bob has the state $\alpha|1\rangle + \beta|0\rangle$ in his possession.
- But he can transform this into the desired state with the application of an X gate.
- How does Bob know what to do? The answer is Alice gives him a call.

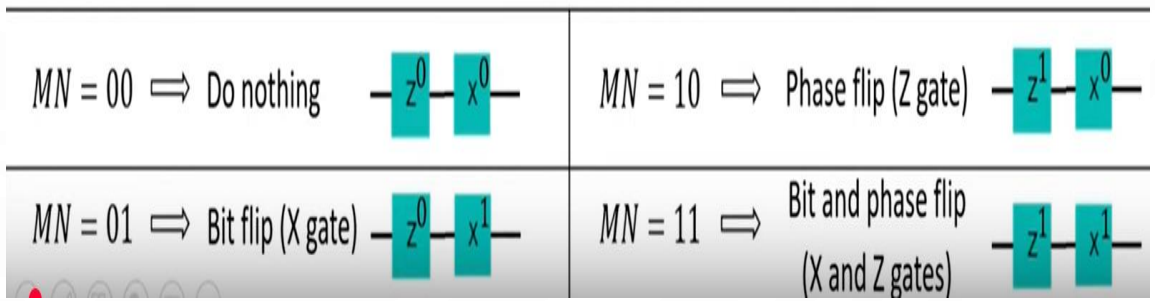
Teleportation step 5: Alice contacts Bob on a classical communications channel and tells him her measurement result

- Alice has to tell Bob her measurement result, and she has to do it using a

classical communications channel-a telephone, email message, radio wave some mechanism governed by the speed of light limit.

- Alice calls Bob and says she got 01, then Bob applies his *X* gate to obtain the state Alice wanted to send to Bob.
- Nothing about that state is communicated over the classical channel—Bob can obtain it because they shared an entangled EPR pair of particles.

Alice's Measurement MN	Bob's State $\|\varphi'\rangle$
00	$\alpha\|0\rangle + \beta\|1\rangle$
01	$\alpha\|1\rangle + \beta\|0\rangle$
10	$\alpha\|0\rangle - \beta\|1\rangle$
11	$\alpha\|1\rangle - \beta\|0\rangle$



DEUTSCH'S ALGORITHM

- Quantum parallelism can be described as the ability to evaluate the function $f(x)$ at many values of x simultaneously.
- One that accepts a single bit as input and produces a single bit as output
- Identity function

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x = 1 \end{cases}$$

- Constant functions

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x = 1 \end{cases}$$

- The identity and bit flip functions are called *balanced* because the outputs

are opposite for half the inputs.

$$f(x) = 0, \qquad f(x) = 1$$

- A function on a single bit can be constant or balanced.
- Deutsch's algorithm will let us put together a state that has all of the output values of the function associated with each input value in a superposition state. Then we will use quantum interference to find out if the given function is constant or balanced.

- The Deutsch–Jozsa problem can be solved efficiently by a quantum computer with no error, whereas a deterministic classical computer would need an exponential number of queries to the black box to solve the problem.
- It yields an oracle that can solve complex problem in polynomial time on a quantum computer.

- Unitary operation U_f that acts on two qubits. It leaves the first qubit alone and produces the exclusive or (denoted by $\oplus$) of the second qubit with the function f evaluated with the first qubit as argument

$$U_f|x, y\rangle = |x, y \otimes f(x)\rangle \qquad x, y \in \{0, 1\}$$

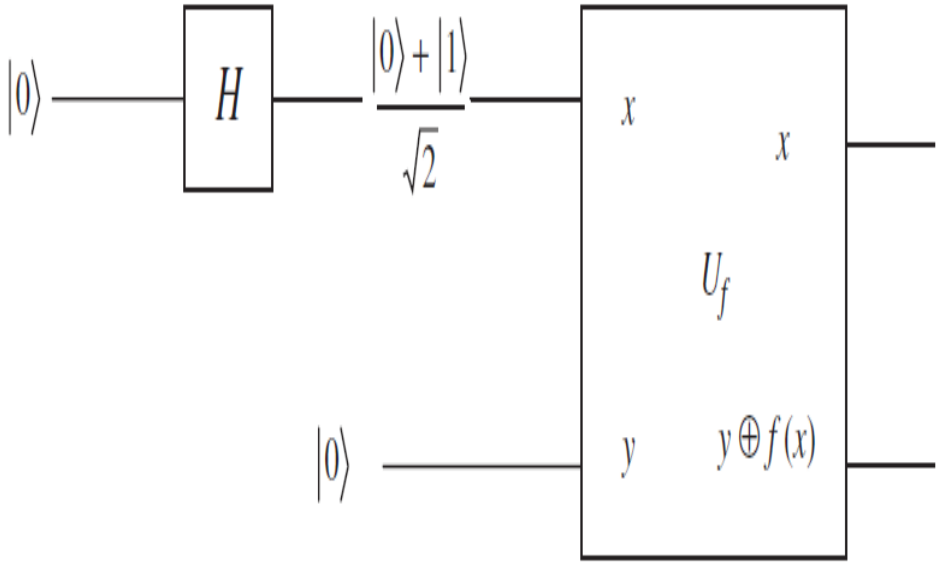
$$U_f\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)|0\rangle = \frac{1}{\sqrt{2}}(U_f|00\rangle + U_f|10\rangle) = \frac{|0, 0 \oplus f(0)\rangle + |1, 0 \oplus f(1)\rangle}{\sqrt{2}}$$

$$0 \oplus 0 = 1 \oplus 1 = 0,\ 0 \oplus 1 = 1 \oplus 0 = 1$$

- If we measure the state $\Sigma |x> |f(x)>$, we get one and only one value of x and $f(x)$. After measurement the system is in a state proportional to $\Sigma x\, f(x)$ for that single and specific value of x.
- Moreover the value of x that we obtain is completely random. So what we have so far is the same as evaluating the function $f(x)$ at some randomly chosen x.

- For a simple function on bits we can learn the value of $f(0)$ or $f(1)$, but not both simultaneously (even though they are simultaneously present in the premeasurement state)
- Deutsch's algorithm takes what we have done so far to exploit the fact that the system is in a superposition state $\Sigma|x\rangle|f(x)\rangle$ to obtain information about a global property of the function, whether or not it is constant $f(0)=f(1)$ or balanced $f(0) \neq f(1)$

- Apply Hadamard gates to the input state |0> |1> to produce a product state of two superpositions.
- Apply U_f to that product state.
- Apply a Hadamard gate to the first qubit leaving the second qubit alone.



$$(H \otimes H)|0\rangle|1\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$
$$= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

$$U_f|x, y\rangle = |x,\ y \oplus f(x)\rangle$$

If $f(0) = 0$:

$$|0, 0 \oplus 0\rangle = |00\rangle.$$

If $f(0) = 1$:

$$|0, 0 \oplus 1\rangle = |01\rangle.$$

In compact notation

$$|0, 0 \oplus f(0)\rangle = (1 - f(0))|00\rangle + f(0)|01\rangle$$

$$U_f|00\rangle = |0, 0 \oplus f(0)\rangle = (1 - f(0))|100\rangle + f(0)|01\rangle$$

$$f(0) = 0 \qquad 0 \oplus f(0) = 0 \oplus 0 = 0$$

$$|0, 0 \oplus f(0)\rangle = (1 - f(0))|00\rangle + f(0)|01\rangle = |00\rangle + (0)|01\rangle = |00\rangle$$

- On the other hand, if $f(0) = 1$, $0 \oplus f(0) = 0 \oplus 1 = 1$

$$|0, 0 \oplus f(0)\rangle = (1 - f(0))|00\rangle + f(0)|01\rangle = (0)|00\rangle + (1)|01\rangle = |01\rangle$$

- Similar logic applied to the other terms gives

$$U_f|01\rangle = |0, 1 \oplus f(0)\rangle = f(0)|00\rangle + (1 - f(0))|01\rangle$$
$$U_f|10\rangle = |0, 1 \oplus f(0)\rangle = (1 - f(1))|00\rangle + f(1)|01\rangle$$
$$U_f|11\rangle = |0, 1 \oplus f(1)\rangle = f(1)|10\rangle + (1 - f(1))|11\rangle$$

$$
\begin{aligned}
|\psi'\rangle &= U_f(H \otimes H)|0\rangle|1\rangle \\
&= (1 - f(0))|00\rangle + f(0)|01\rangle + f(0)|00\rangle + (1 - f(0))|01\rangle \\
&\quad +(1 - f(1))|00\rangle + f(1)|01\rangle + f(1)|10\rangle + (1 - f(1))|11\rangle
\end{aligned}
$$

- The Hadamard gate is applied to the first qubit, and the second qubit is left alone

$$
|\psi_{out}\rangle = (H \otimes I)U_f(H \otimes H)|0\rangle|1\rangle
$$

- First couple of terms explicitly

$$
\begin{aligned}
&(H \otimes I)[(1 - f(0))|00\rangle + f(0)|01\rangle] \\
&\quad = (1 - f(0))(H|0\rangle) \otimes |0\rangle + f(0)(H|0\rangle) \otimes |1\rangle \\
&\quad = (1 - f(0)) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes |0\rangle + f(0) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes |1\rangle \\
&\quad = (1 - f(0)) \left(\frac{|00\rangle + |10\rangle}{\sqrt{2}} \right) + f(0) \left(\frac{|01\rangle + |11\rangle}{\sqrt{2}} \right)
\end{aligned}
$$

- Final output state of Deutsch's algorithm as

$$|\psi_{out}\rangle = (1 - f(0) - f(1))|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) + (f(1) - f(0))|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

- Now suppose that the function is constant so that f(0)=f(1)

$$|\psi_{out}\rangle = -|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

- On the other hand, if $f(0) \neq f(1)$.

$$|\psi_{out}\rangle = \pm|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \qquad (f(0) \neq f(1))$$

- Classically: To know both f(0) and f(1), you must feed the oracle twice: once with input 0, once with input 1.
- Quantum Mechanically: We can put the input qubit in a superposition of 0 and 1, and then query the oracle only once. Because the oracle is unitary, it acts linearly on all components of the superposition, effectively evaluating f(0) and f(1) "in parallel.“
- In one application of U_f, the quantum state has “experienced” both f(0) and f(1).
- Both computations coexist in superposition.
- However we can’t just measure and read out both values at once (measurement collapses to one outcome). Instead, the trick is to use interference (Hadamard on the first qubit) to extract a global property of the function (whether it’s constant or balanced).
- After the oracle, applying a Hadamard gate on the first qubit causes the amplitudes corresponding to f(0) and f(1) to interfere.

Important video links

https://www.youtube.com/watch?v=B3U1NDUiwSA

https://www.youtube.com/watch?v=PH1FbkLVJU4

https://www.youtube.com/watch?v=pcGlBacW-q0&t=517s

https://www.youtube.com/watch?v=mAgnlj0UXLY